

Visão Artificial e Realidade Mista

Computer Vision and Mixed Reality

Mestrado em Engenharia Informática e Multimédia

Master in Informatics and Multimedia Engineering

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Pedro Mendes Jorge

pedro.mendes.jorge@isel.pt / Room F.0.29

Goals

- Understand the concepts of Augmented Reality (AR), Augmented Virtuality (AV) and Virtual Reality (VR) - *Mixed Reality/Virtuality Continuum*
- Know examples and applications involving the concepts of AR/AV/VR
- Learn Computer Vision (CV) concepts for augmented reality
- Know devices and technologies to implement examples of augmented reality

Syllabus – Theoretical Component

- Chapter 1. Introduction to Augmented Reality
- Chapter 2. Displays
- Chapter 3. Tracking
- Chapter 4. Computer Vision for Augmented Reality
- Chapter 5. Calibration and Registration

Practical Component

- Implement 2 practical examples of augmented reality
 - Project 1 - Computer Vision based
 - Project 2 - Integrating AR application
- Devices: cameras mono&stereo, RGB-D (RealSense); LeapMotion; smartphone; Magic Leap; Meta Quest 3.
- Technologies: OpenCV, Unity, ARToolkit, Vuforia, ARCore, ARKit.

Evaluation

- Final Grade = 50% P1 + 50% P2
- Each **individual project** includes the development of an application and a written report with final discussion.

Bibliography

- Augmented Reality: Principles and Practice, 1st Edition by Dieter Schmalstieg and Tobias Hollerer, June 2016, Pearson Education
- Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR, Steve Aukstakalnis, 2017, Addison-Wesley
- Augmented Reality for Developers, by Jonathan Linowes and Krystian Babilinski, October 2017, Packt Publishing Limited

Quote

There are two types of students:

Those who use AI to learn more and those who use AI to learn less!

@Mustafa Suleyman (CEO, @MicrosoftAI | Author: The Coming Wave)